

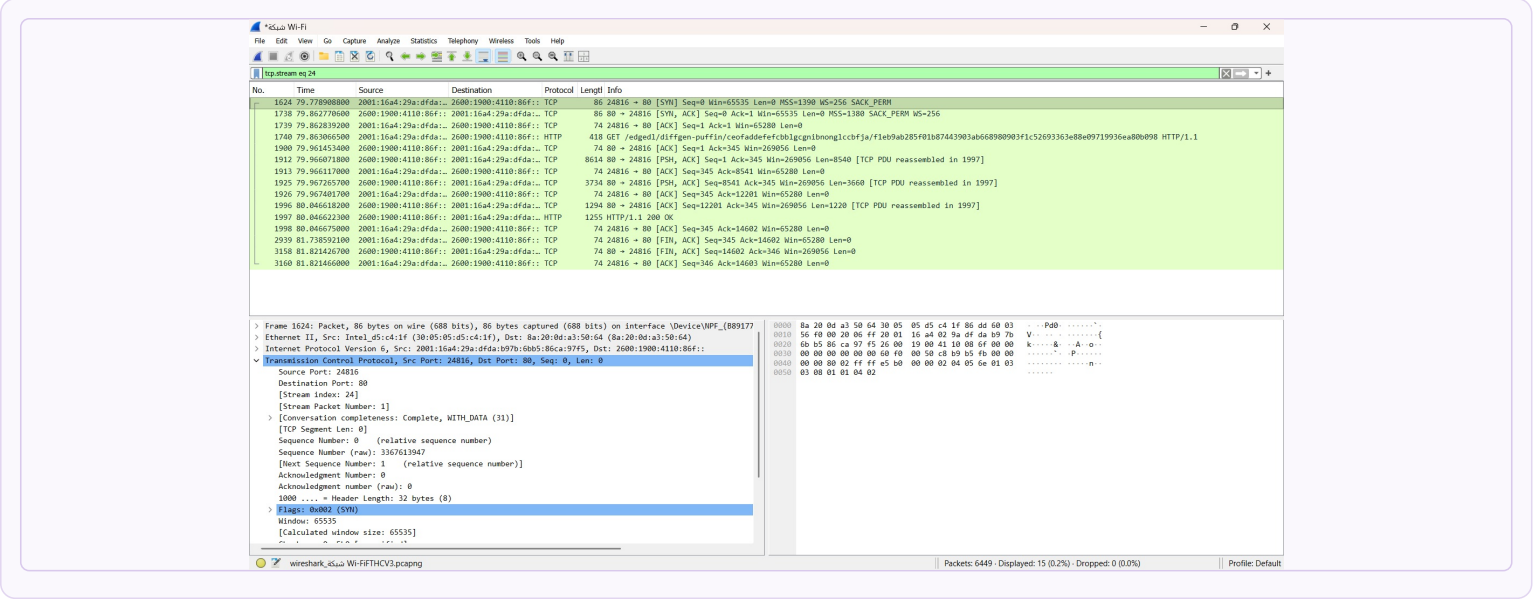
Lab 1: The Internet Protocols

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Course: CS471 — Web Technologies

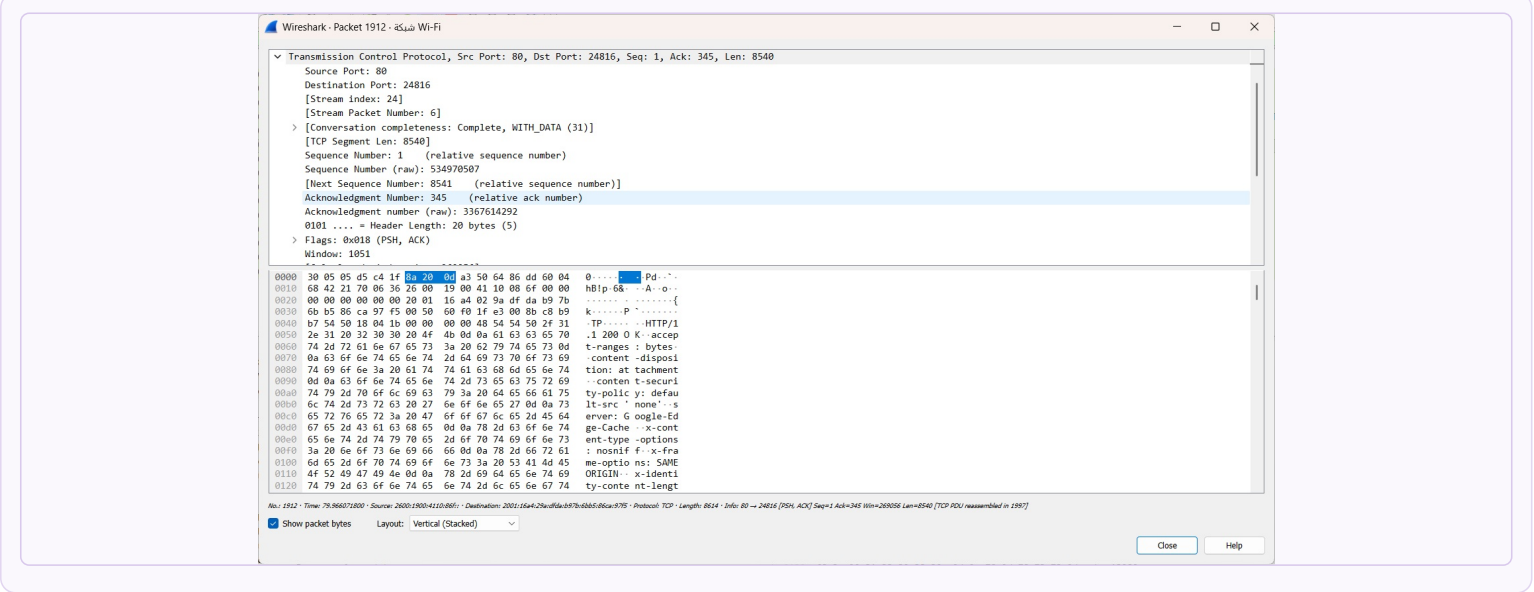
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Screenshot 1 — TCP Three-Way Handshake



TCP Three-Way Handshake (SYN, SYN-ACK, ACK)

Screenshot 2 — TCP Data Transfer



TCP Data Transfer showing sequence and acknowledgment information

Table 1 — TCP or UDP and Reasons

Criteria	TCP or UDP	Reasons
Reliability and Connection Establishment	TCP	TCP establishes a connection before data transfer and provides reliable delivery.
Data Integrity and Ordering	TCP	TCP uses sequence and acknowledgment numbers to ensure data arrives correctly and in order.

Table 2 — Use Cases and Performance

	TCP	UDP
Use Cases	Web browsing, email, file transfer	Streaming, VoIP, online gaming, DNS
Performance	Slower because of connection establishment and reliability mechanisms	Faster with lower overhead because it does not establish a connection